Advanced Machine Learning, ETH Zurich

Hinge-fan networks

One hidden layer, one hinge per kink

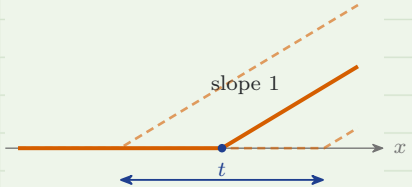
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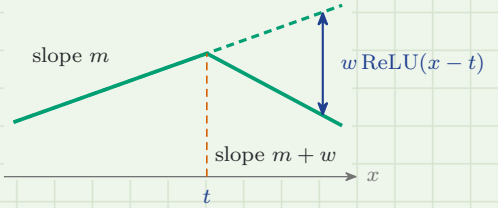
Entry 1

One layer of hinges

A hinge is the ReLU of a shifted input, $\mathrm{ReLU}(x - t)$: it stays at 0 left of t and rises with slope 1 right of it. Its position t is the only number it has.



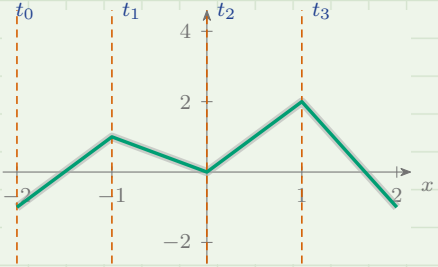
Solid: the hinge $\mathrm{ReLU}(x - t)$. Dashed: the same hinge at two other positions. Changing t slides it along x and keeps its shape.



Add $w\,\mathrm{ReLU}(x - t)$ to a curve of slope m (dashed): from t on, its slope is $m + w$ (solid).

The weight w turns the curve at t, up for $w > 0$ and down for $w < 0$. In a network, t sits in the hinge and w on the arrow that leaves it.

Start from the constant c. A hinge at the left end a is a line on $[a, b]$, so it sets the first slope; every further hinge turns the curve at its position t_j by its weight w_j.



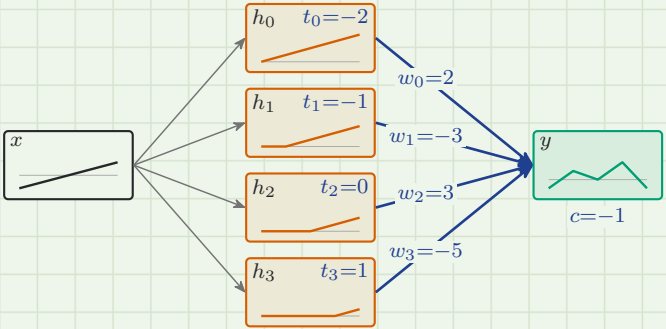
Grey: the zigzag f. Green: the sum so far, one hinge more per click.

$$y = c + \sum_j w_j \,\mathrm{ReLU}(x - t_j)$$

$$\begin{aligned} y = {} & -1 && \text{the bias } c = f(-2) \\ & +2\,\mathrm{ReLU}(x+2) && t_0 = -2, \text{ slope } 0 \text{ to } 2 \\ & -3\,\mathrm{ReLU}(x+1) && t_1 = -1, \text{ slope } 2 \text{ to } -1 \\ & +3\,\mathrm{ReLU}(x) && t_2 = 0, \text{ slope } -1 \text{ to } 2 \\ & -5\,\mathrm{ReLU}(x-1) && t_3 = 1, \text{ slope } 2 \text{ to } -3 \end{aligned}$$

On $[-2, 2]$, $\mathrm{ReLU}(x + 2) = x + 2$: hinge 0 sits at the left end and is a line, so with c it gives the first piece, $2x + 3$.

Hinge j reads x with the fixed weight 1 and bends at its position t_j. The output y adds the hinges up, with the weights w_j and the bias c.



Every box draws its neuron on $[-2, 2]$. The hinges, on one scale, are one ramp slid to -2, -1, 0, 1.

The output y is the zigzag.

$$h_j = \mathrm{ReLU}(x - t_j)$$

$$y = c + \textstyle\sum_j w_j h_j$$

Free, in pen: t_j and w_j for every hinge, and c: $2n + 1$ numbers, nine here. A grey arrow has the weight 1. The hinges can come in any order.

Entry 2

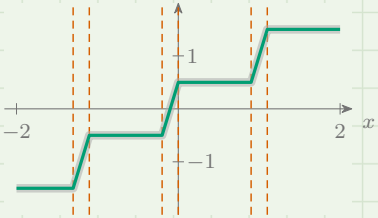
The construction

Read the numbers off f: the bias is its value at a, hinge 0 sits at a with the first slope m_0, and hinge i sits at the kink t_i with the change of slope $\Delta_i = m_i - m_{i-1}$ there.

output	$c = f(a)$		
hinge 0	$t_0 = a$,	$w_0 = m_0$	
hinge i	t_i, its kink,	$w_i = \Delta_i$	$i = 1, \dots, K$

Then $y = f(a) + m_0 \,\mathrm{ReLU}(x - a) + \sum_i \Delta_i \,\mathrm{ReLU}(x - t_i)$, and $y = f$: both are piecewise linear, they agree at a, and on the ith piece both have the slope $m_0 + \Delta_1 + \cdots + \Delta_i = m_i$.

K kinks take $K + 1$ hinges, and K when f starts flat: then $m_0 = 0$ and hinge 0 goes.



The staircase of the page starts flat: six kinks, six hinges, $c = -1.5$ and $w_j = 5, -5, 5, -5, 5, -5$.

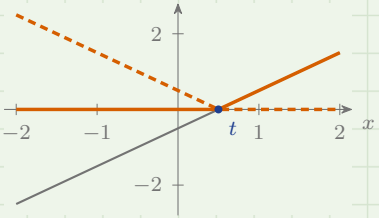
A hidden neuron usually has two numbers, $\mathrm{ReLU}(ux+v)$, and bends at $t=-v/u$. On $[a,b]$ each such neuron is a hinge of ours with a weight, plus at most a line and a constant.

$u>0$: $\mathrm{ReLU}(ux+v)=u\,\mathrm{ReLU}(x-t)$.
u moves onto the outgoing weight:
$\mathrm{ReLU}(\lambda z)=\lambda\,\mathrm{ReLU}(z)$ for $\lambda\geq 0$.

$u<0$: $\mathrm{ReLU}(ux+v)=|u|\,\mathrm{ReLU}(t-x)$, which opens to the left.
$\mathrm{ReLU}(t-x)=\mathrm{ReLU}(x-t)-(x-t)$, since $\mathrm{ReLU}(z)-\mathrm{ReLU}(-z)=z$, and on $[a,b]$ the line is $x-t=\mathrm{ReLU}(x-a)+(a-t)$.

$u=0$: a constant, which c takes.

So n arbitrary neurons make at most $n+1$ of our hinges: the one at a collects every line.



Dashed: $\mathrm{ReLU}(t-x)$. Solid: ours, $\mathrm{ReLU}(x-t)$. Grey: the line $x-t$, solid minus dashed.

Entry 3

Fitting

Fix the positions $t_1, \dots, t_n$. Then $y = c + \sum_j w_j \varphi_j(x)$, with the features $\varphi_j(x) = \mathrm{ReLU}(x - t_j)$, is linear in the numbers $\theta = (c, w_1, \dots, w_n)$ that are left.

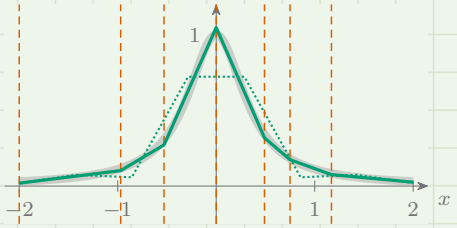
On the points $x_1, \dots, x_N$, with f the target's values there, the mean squared error $\frac{1}{N}\|\Phi\theta - f\|^2$ is a convex quadratic in θ. Row i of the matrix Φ holds the features at x_i:

$$\big(1,\ \varphi_1(x_i),\ \dots,\ \varphi_n(x_i)\big).$$

Its minimizers solve the normal equations

$$\Phi^\top \Phi\, \theta = \Phi^\top f$$

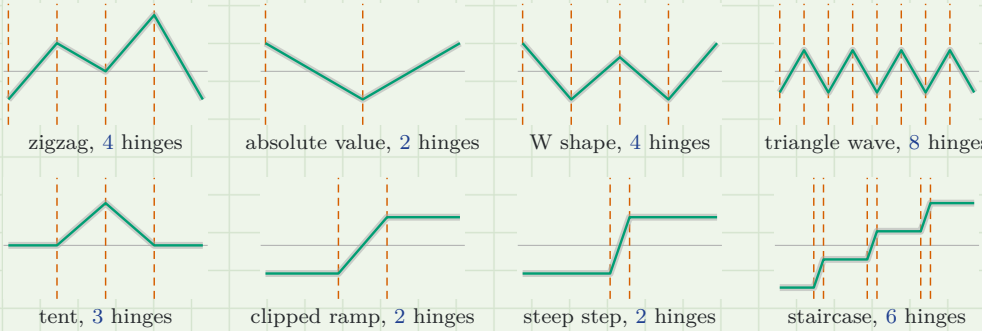
The hidden layer picks the features, and the output is linear regression on them, as in the course; random features pick the hidden layer at random. Placing the hinges is the nonconvex part.



Seven hinges spread evenly from -2 straddle the peak, and the best weights miss it: MSE 8.0×10^{-3}.

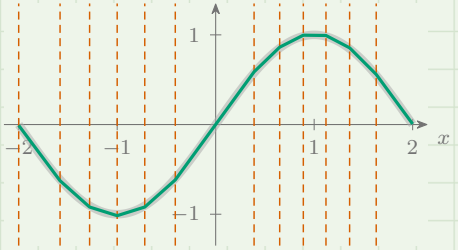
Placed by the fit of the next frame, one hinge sits on the peak: MSE 1.3×10^{-4}, about sixty times lower. Dotted: the even fit.

The page's fit places the hinges by a dynamic program, with a hinge at a and without, sets c and the w_j by least squares, and keeps the smaller error.



Grey: the target, under the fit; dashed: the hinges. Top row: one hinge per kink, and one for the first slope. Bottom row: these start flat, so one per kink is enough. No fit with one hinge fewer is exact.

If $|f''| \le M$, the interpolant through $n+1$ evenly spaced points is a network of n hinges, and it stays within $2M/n^2$ of f on $[-2, 2]$.



The sine $\sin(\pi x/2)$ with 12 hinges: MSE 3.3×10^{-5}, largest error 0.017.

Each doubling of the hinges divides the largest error by about 4, and the MSE by about 16.

Hinges until the largest error drops below 0.02, about two pixels on the page:

parabola	9	sine	11
Runge function	9	cubic	11
Gaussian bump	11	two periods	21
		chirp	24

Least squares beats interpolating f at the same kinks: with $n = 10$ it lowers the MSE about 4 to 6 times, for the sine to 7.0×10^{-5} from 3.6×10^{-4}.

Try them on the page: eth-aml-hs26.github.io/neural-networks/viz/hinge-fan-networks

Entry 4

Summary

The hinge chain and the hinge fan both build a piecewise-linear function out of hinges. They differ in depth, in width, and in what each hinge reads.

	Hinge chain	**Hinge fan**
Hidden layers	K	1
Neurons per layer	2	$K+1$ (K if f starts flat)
A hinge reads	the hinge before it	x
ReLU on	every neuron, lift C	the hinges
Free numbers	gaps d_ℓ, slope changes Δ_ℓ	t_j, w_j; c
The kinks	in order, by the gaps	anywhere
Zigzag: neurons, free numbers	6, 7	4, 9

Each network has its page, where you pick a target and set every free number by hand or let fit set them:

Hinge-fan networks `eth-aml-hs26.github.io/neural-networks/viz/hinge-fan-networks`
Hinge-chain networks `eth-aml-hs26.github.io/neural-networks/viz/hinge-chain-networks`